1. Data Collection

Data collection is the first step in machine learning. In this step, data is gathered from different sources such as databases, surveys, websites, or files. For this project, the student performance dataset was created and stored in a CSV file. The dataset contains Study_Hours, Attendance, Assignments, and Marks. Good quality data helps build an accurate machine learning model.

2. Data Cleaning

Data cleaning is the process of checking and fixing errors in the dataset. It includes handling missing values, removing duplicate records, and correcting incorrect data. In this dataset, there are no missing values or duplicates. All columns contain numerical values and are already clean. Therefore, the dataset is ready for analysis and model training.

3. Feature Selection

Feature selection means choosing the input variables that will be used to train the model. In this project, Study_Hours, Attendance, and Assignments are selected as features. These factors can influence student marks. Selecting relevant features helps improve model performance and prediction accuracy.

4. Train-Test Split

The dataset is divided into training data and testing data. Training data is used to teach the model, while testing data is used to evaluate it. In this project, 80% of the data is used for training and 20% for testing. This helps measure how well the model performs on unseen data.

5. Model Training

Model training is the process of teaching the machine learning algorithm using the training data. In this project, the Linear Regression algorithm is used. The model learns the relationship between Study_Hours, Attendance, Assignments, and Marks. After training, the model can make predictions for new data.

6. Testing

Testing is performed using the test dataset. The trained model predicts marks for the testing data. These predictions are then compared with the actual marks. Testing helps determine how accurately the model works on unseen data.

7. Evaluation

Evaluation measures the performance of the model using metrics such as Mean Absolute Error (MAE) and R^2 Score. MAE measures the average prediction error, while R^2 Score indicates how well the model explains the data. In this project, the model achieved a low MAE and a high R^2 Score, indicating good performance.

8. Prediction

Prediction is the final step of the machine learning workflow. After training and evaluation, the model is used to predict marks for new students. The model uses Study_Hours, Attendance, and Assignments as inputs and predicts the expected marks. This helps estimate student performance based on available information.